

Raj Khatri

LinkedIn: <https://linkedin.com/in/rajkhatri2295> · **Portfolio:** <https://rajkhatri08.github.io>
GitHub: <https://github.com/rajkhatri08>

Mobile: +91 7252091514
Email: rajkhatrigbpuat@gmail.com

PROFESSIONAL SUMMARY

Final-year Industrial & Production Engineering student working at the intersection of plant diagnostics and applied GenAI. Built and deployed a Plant Health & Root-Cause Agent that detects chemical-plant faults early and proposes an evidence-based root cause for a supervisor to approve, evaluated once on a sealed test set. At Tata Motors, diagnosed a live Body-in-White defect as an upstream supplier-side cause rather than on-line equipment wear, using regression-based trend analysis confirmed by physical measurement on the shop floor. Reads plant process reasoning and writes Python.

EDUCATION

B.Tech, Industrial & Production Engineering
College of Technology, G.B. Pant University of Agriculture & Technology, Pantnagar

Uttarakhand, India
August 2023 – June 2027

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL / MySQL), JavaScript, Bash

GenAI & ML: LLM application development (Gemini, LangGraph), prompt engineering & constraint design, RAG pipelines, agentic tool routing, vector search (ChromaDB, pgvector), embeddings (MiniLM / ONNX), grounded generation & decline behaviour, PCA fault detection (T^2 / SPE), Scikit-learn

Backend: FastAPI, Pydantic data contracts, SQLAlchemy, PostgreSQL, pgvector, star-schema modelling, Flask, Next.js, Supabase, REST API design, pytest, GitHub Actions CI

Data & Tools: Pandas, NumPy, SciPy, Matplotlib, Seaborn, Jupyter, Excel (Advanced / VBA), Power BI, Git/GitHub, Render, Vercel, Neon

Domain: Root-cause analysis, upstream/downstream fault isolation, DFMEA, Six Sigma, statistical modeling & hypothesis testing, pre-registered evaluation-set design

EXPERIENCE

Tata Motors
Industrial Trainee — Data Analytics
Project: "Predictive Quality Intelligence — A Data-Driven Framework for BIW Root-Cause Diagnosis"

Pantnagar, Uttarakhand (**On-site**)
03 July 2026 – 03 August 2026

- Diagnosed a recurring Body-in-White quality failure as an upstream supplier-side dimensional defect, ruling out on-line equipment wear.
- Applied linear regression with R^2 confidence scoring across ~300 inspection points, then validated it by physical measurement on the shop floor.
- Designed a containment procedure deployable without production stoppage or vendor dependency.
- Built an interactive diagnostic dashboard tracking defect trends across 10 monitored quality metrics for ongoing condition review.
- Authored a 20+ section technical report benchmarking the findings against DFMEA, Six Sigma, and peer-reviewed manufacturing literature.

Voldebug Innovations Pvt. Ltd.
Backend Developer (Part-time)

Remote
22 August 2026 – Present

- Migrating a live K–12 AI-literacy platform from browser-local persistence to a Supabase (PostgreSQL) backend — schema design, row-level security policies, and role-based authentication for student, teacher, and principal accounts.
- Implementing server-side Next.js APIs for XP/badge progression, class-code enrollment and teacher analytics, replacing client-only state.
- Establishing database migrations, staging/production environment separation, and a repeatable deployment workflow on Vercel.

Intern

01 July 2026 – 31 July 2026

- Built the platform's first working version — a gamified AI-literacy SPA spanning student, teacher, and principal roles, with a curriculum-linked prompt library and an XP/badge/streak progression system.
- Authored the phased Supabase + Next.js migration plan now being implemented in the backend role above.

PROJECTS

Plant Health & Root-Cause Agent *Python, scikit-learn, LangGraph, Gemini, FastAPI*

Sep–Oct 2026 · **Live demo**

- Built an end-to-end plant-health system on the Tennessee Eastman chemical-plant benchmark: PCA fault detection calibrated to ~1 false alert/day, equipment-level attribution, and a governed library of 12 failure modes.
- Added a LangGraph agent in which one Gemini call explains the matcher's pick, a deterministic check verifies every cited fact, and a supervisor approves; passed every zero-tolerance case in a 36-case safety set.
- Pre-registered the evaluation and ran it once on a sealed test set: 95.6% of fault runs detected (94.8% on unseen fault types), 1 alert per fault episode vs 25–35 from per-tag alarms, and the right cause ranked first 75.7% of the time.
- Reported where the LLM hurt (–3.8 pts top-1) instead of hiding it; 2,253 tests in CI; Rs 0.05 per diagnosis.

Supplier Quality Risk Agent *Python, FastAPI, Gemini, Postgres + pgvector, Pydantic · Render / Vercel / Neon*

Live demo

- Built a dual-source AI agent that decides per query whether an answer needs structured supplier data, corrective-action documents, both, or neither — declining when no source supports one — with 88–92% routing accuracy across five runs on a pre-registered 24-question set.
- Replaced text-to-SQL with a fixed catalogue of eight parameterised queries, trading question coverage for reviewability in a system where a wrong number triggers a supplier corrective action.
- Caught a regression where a prompt improvement made the router fabricate evidence for a non-existent supplier; constrained the prompt with the real entity list, raising decline accuracy to 5/5.

Source-Governed RAG *Python, FastAPI, ChromaDB, MiniLM/ONNX, PostgreSQL, Next.js · Render / Vercel / Neon*

Live demo

- Built an end-to-end RAG system over regulatory documents where a governance layer vetoes untrustworthy sources, with approval status and review dates overriding similarity ranking.
- Improved retrieval accuracy from 78% to 95% on a labelled evaluation set by replacing fixed-size chunking with document-structure chunking.